

# Summer Internship

## June-July 2025

### CoDMAV Review 3

Project Title : Credit Risk Analysis using a Post Quantum secure Federated Learning approach

Project ID : 10

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Deliverable Submitted: IEEE paper and a Proof of concept

# Outline

- Abstract
- Problem Statement and Research Gap
- Literature Survey( At least 5 recent papers)
- Suggestions from review-2
- Objectives
- Design(Architecture/Flowchart/use case diagram/or
- Class Diagram)
- Methodology Adopted- Algorithms/Techniques Adopted
- Implementation- Code snippets
- Results – output screenshots
- Conclusion and Future Enhancement
- References

# Abstract

Protecting sensitive consumer data has become a top priority in today's digital banking environment, particularly as organizations use AI-driven models for loan decisions, fraud detection, and credit risk assessment. The widely used traditional encryption techniques, such as RSA, Elliptic Curve Cryptography (ECC), and Diffie-Hellman, are extremely susceptible to upcoming quantum attacks, especially because of Shor's algorithm, which can crack these schemes in polynomial time.

We suggest a Decentralized Post-Quantum Cryptography-enhanced Federated Learning (PQC-FL) framework to future-proof banking systems. Our system incorporates a hybrid asymmetric cryptographic scheme that combines AES (for effective symmetric encryption), Dilithium and Falcon (for quantum-safe digital signatures), and Kyber (for key exchange). In order to provide quantum-resilient confidentiality, integrity, and authentication, these cryptographic standards were chosen in accordance with NIST's post-quantum cryptography recommendations.

Our method uses a decentralized federated architecture with three separate servers that cooperatively coordinate model training, in contrast to conventional FL setups that depend on a single server. By preventing a single point of failure, this design improves fault tolerance and learning process trust. Every banking client shares only encrypted model updates and trains a local model on non-IID, frequently unbalanced, customer data. These updates are compiled by the decentralized servers using safe protocols that preserve verifiability and privacy. To deal with data imbalance and heterogeneity, sophisticated methods like adaptive client participation, class-weighted loss, and oversampling are used. Our findings show that this decentralized, quantum-secure system makes it possible to deploy AI in banking settings in a way that protects privacy, complies with regulations, and is prepared for the future.

# Problem statement and Research gaps Identified

So the problems that we've come across in existing literature and projects were the following :

1. **Data privacy** : Data privacy threats in centralised repositories for traditional ML techniques have caused a serious issue for implementing AI features into sensitive sectors like banking and healthcare. With the decentralised feature of FL we are able to train models locally on the edge devices repo rather than on the central database, exposing it to threats.
2. **Security gaps in FL** : In our process to evaluate the privacy given by FL we've noticed a huge security threat namely wherein the parameters being passed back and forth between the client and the server, they are at risk of being eavesdropped or may even accept parameters from malicious clients. Hence with a stable encryption for validation and confidentiality we aim to solve this problem.
3. **Tradeoffs seen in an PQC FL framework**: We had speculated tradeoffs seen in the PQC FL framework mainly with security improvements with various other parameters like accuracy, space used, time overhead, and aggregation quality.  
We later came up with cases to slowly test the hypothesis from the upper bound to the real life scenarios on how the FL model would work

# Literature Survey

| Approach/Paper                                                         | Primary Focus                                                                          | Security Method                            | Key Limitation                                                                                                   | What we aim for                                                                                                              |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Wang et al.</b><br><u><a href="#">cite_start</a></u><br>"FedCredit" | Applying standard federated learning to credit risk domain.                            | Classical Cryptography (eg RSA,ECC)        | The framework is vulnerable to attacks from future quantum computers that can break classical encryption.        | We integrate Post-Quantum Cryptography (PQC) to ensure the system is secure against both current and future quantum threats. |
| <b>Commey et al.</b><br><u><a href="#">cite_start</a></u><br>"PQS-BFL" | Using a Blockchain for decentralized authentication and auditability in PQC enabled FL | PQC Signatures for on-chain authentication | The focus is on authentication and ledger integrity, not the confidentiality of the model parameters themselves. | We provide end to end security (confidentiality of a high level with verification using similar to smart contracts)          |

# Literature Survey

| Approach/Paper                                                                 | Primary Focus                                                                   | Security Method                                                                                         | Key Limitation                                                                                                               | What we aim for                                                                                                            |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Li, P., et al. <a href="#">cite_start</a><br>- "Enhancing Quantum Security..." | Integrating PQC directly to secure the general FI process                       | PQC Cryptography often with a focus on signatures rather than key encapsulation methods (KEMS)          | This paper acknowledged the rather exponential increase in time by 10% for every 10 clients reaching 3 hours for 100 clients | We plan on using the symmetric and an asymmetric approach dynamically to solve for the time constraints.                   |
| <b>Li, et al. (2024)</b><br><i>"Enhancing Quantum Security"</i>                | Establishing the foundational framework for enhancing efficiency seen in PQC FL | Have done a comparative study on what PQC algorithms achieved and what aggregate strategy proved useful | A brute force symmetric approach caused much loss and could be improved by adding a dynamic solution                         | An asymmetric approach wherein the protocol is chosen based on load to encrypt / decrypt and dynamically shifting to that. |

# Objectives

1. Integrate and Benchmark  
Develop a robust PQC enabled FL framework using flower and liboqs and establish baseline performance metrics for a realist credit risk use case.
2. Design and Implement Asymmetric Security :  
Implement an asymmetric communication protocol where security requirements can be varied (eg lightweight client to server , heavyweight server to client) to optimize resource usage and reduce overhead,
3. Quantify and Analyze the tradeoffs:  
Systematically measure and plot the impact of different PQC schemes (symmetric vs asymmetric) on key metrics model accuracy, training time ,communication time, aggregation quality,particularly in an non IID environment

These were the baseline Objectives that we had and these showed the future scope of the project to analyze the effects of the fit parameter on accuracy, the number of clients versus the training time, the resources used for classical vs PQC approach.

# Methodology adopted

To effectively see the tradeoffs that we have hypothesized between security and a multitude of other features , we have developed a few cases that start from an ideal pure FL to the realistic scenario that we hope to address. Here are few of the scenarios :

1. Pure FL with IID data
2. FL - IID Data with PQC overhead
3. FL with Non IID data with Classical schemes
4. FL - NIID data with Asymmetric PQC

These four cases that we decided to go over shows first in the ideal scenario what the upper bound on resource management, accuracy and aggregation quality would look for our experiment , and as per our hypothesis this would vary in multiple ways with each offering its own pros and cons.

We use flower for FL orchestration, liboqs for NIST standardized PQC algorithms, an indian credit risk dataset with non IID partitioning.

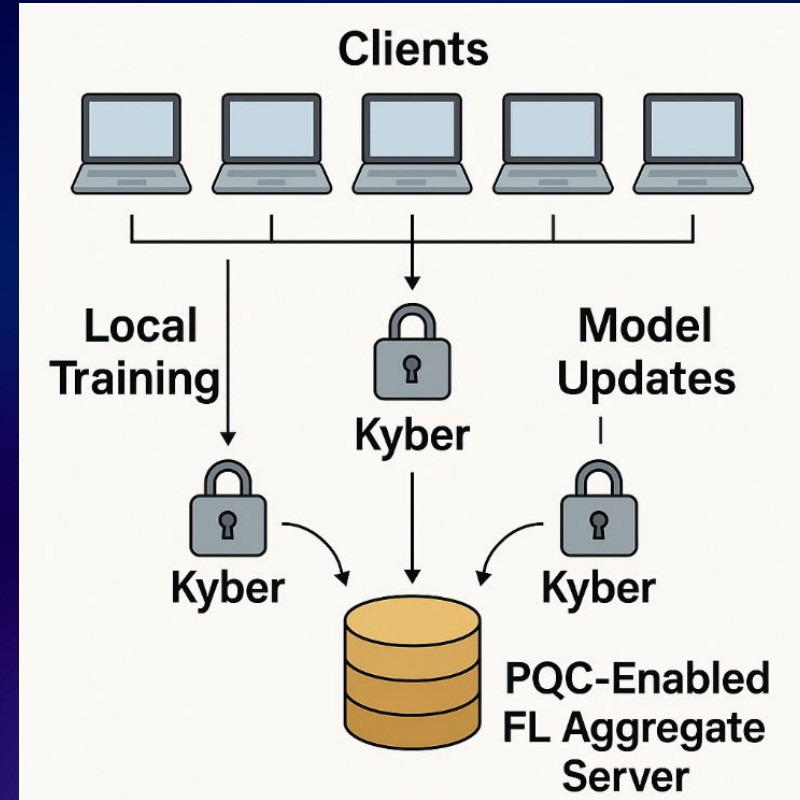
50 clients to simulate a real world consortium of financial institutions.



# Design

So in a traditional FL system , this is how it would look like, Multiple edge devices or as we call them the local repos and the FL simulation is started off with the server passing an initial set of parameters to the clients for them to begin training on.

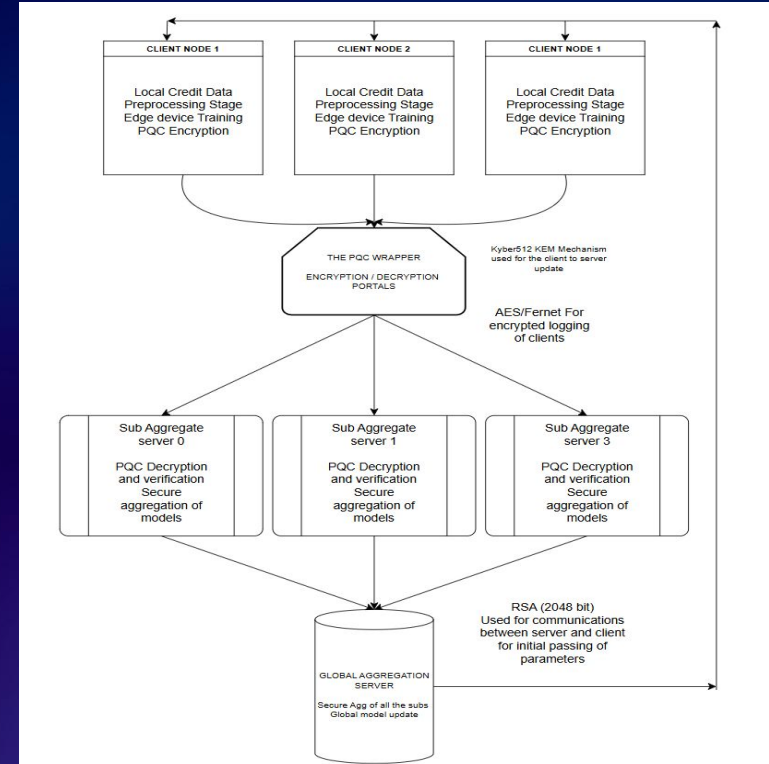
1. **Initialize Global Model:** The central server starts with an initial machine learning model.
2. **Distribute Model:** The server sends the current global model to selected clients.
3. **Local Training:** Each client trains the model on its private local data.
4. **Send Updates:** Clients send their model updates back to the server.
5. **Aggregate Updates:** The server combines the updates to improve the global model.
6. **Repeat:** These steps are repeated until the model performs acceptably.



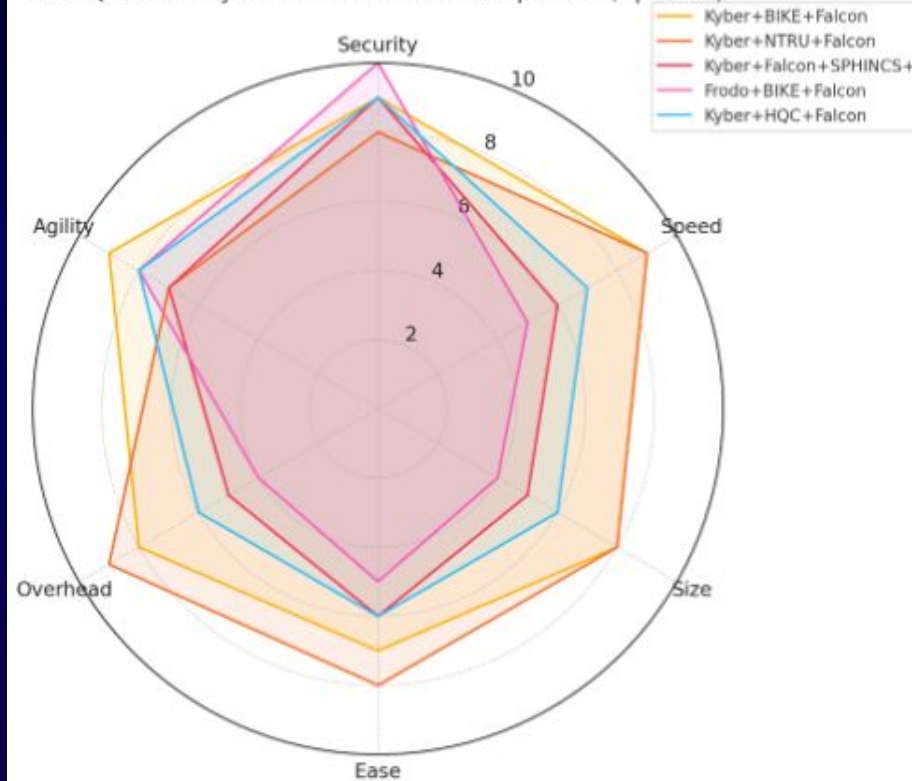
# Design

Now in our setup the PQC wrapper comes between the parameter passing and offers the encryption as a service to the outgoing and incoming parameters and uses the Kyber512 PQC algorithm to securely encrypt from the client to the server and then waits for all the clients to complete training , aggregates them basically a weighted average and then stores this and passes this new average parameter onto the next round.

This adds the PQC overhead to computational time, resources used and may even affect accuracy. We have chosen Dilithium and Falcon as our two primary algorithms as they have a given average over all the parameters measured like signature size , security , ease and overhead as the parameters taken into consideration.



Post-Quantum Hybrid Scheme Metric Comparison (Updated)



# Implementation

## Federated Learning Orchestration:

- **Flower (flwr):** Used to manage the entire simulation, including client selection, communication rounds, and server-side aggregation. We implemented a custom `Strategy` to intercept model updates for security processing.

## Post-Quantum Cryptography Layer:

- **Liboqs-python:** Provides direct access to C implementations of PQC algorithms.
  - **Key Encapsulation (Confidentiality):** Kyber512 is used for securely establishing a shared secret.
  - **Digital Signatures (Authenticity):** Dilithium and Falcon-512 are used to sign and verify model updates.
- **Cryptography library:** Provides fast, standardized **AES encryption** (via Fernet) using the shared secret established by Kyber.

## Machine Learning & Data Processing:

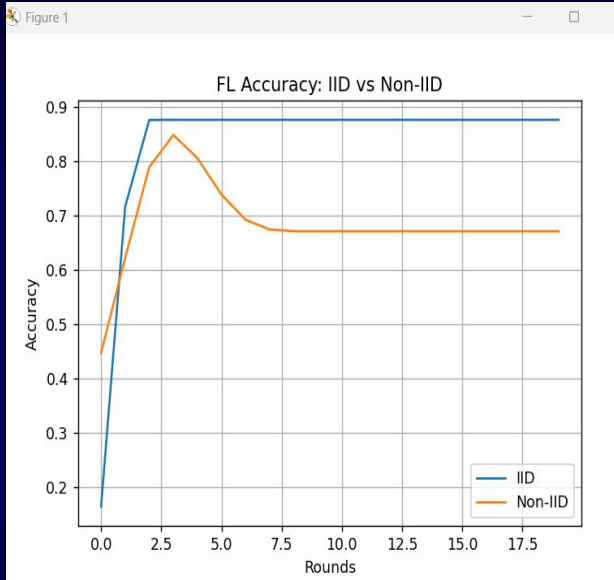
- **PyTorch:** For defining, training, and evaluating the neural network model.
- **Pandas & Scikit-learn:** For loading the dataset partitions and performing label encoding.

# Challenges Faced

During the completion of the review 2 deliverables we found a variety of errors and improvements that we could make to our final presentation on the third review

1. The Ray implementation of flower – Flower is the module that we were using in python to simulate our FL system, however that was turning out to be a disaster as it kept crashing due to internal conflicts and prompted us to look forward to contributing on its open source to enable a PQC FL system directly through a whole new framework within flower.
2. The Non IID partitioning of the data has led the clients to all have strict partition of data, having all from the same cities and some without frauds etc. Clearly the Preprocessing stage had to come back and we've tweaked it to allow for a much better partitioning of data.
3. Serialization and deserialization of the parameters – These repeatedly kept us from progressing as it had few compatible data types to which the parameters in byte form could we be converted to for our encryption purposes , finally we set on the `numpy.array()` type.

# RESULTS



**1. Data Processing: Prudent Data Division:** Even with non-IID data, significant performance drops can be avoided if the partitioning makes sure that each client receives a sufficiently diverse subset of the total data

**Undersampling and Oversampling:** we have implied technique of undersampling . By balancing the local datasets, this facilitates the learning process for each client and, in turn, for the global model.

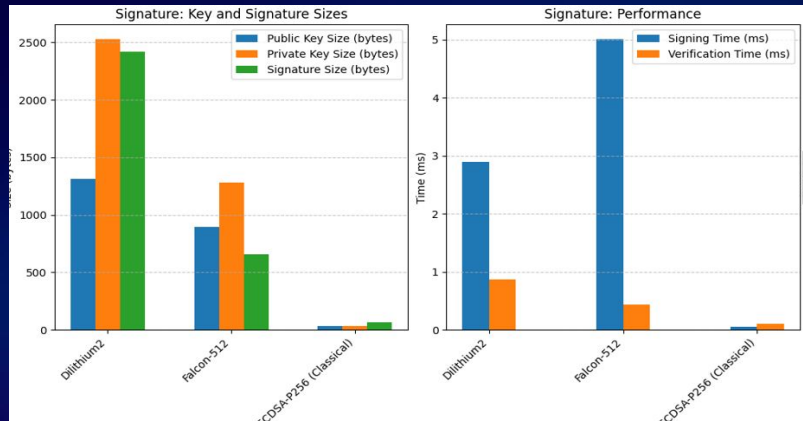
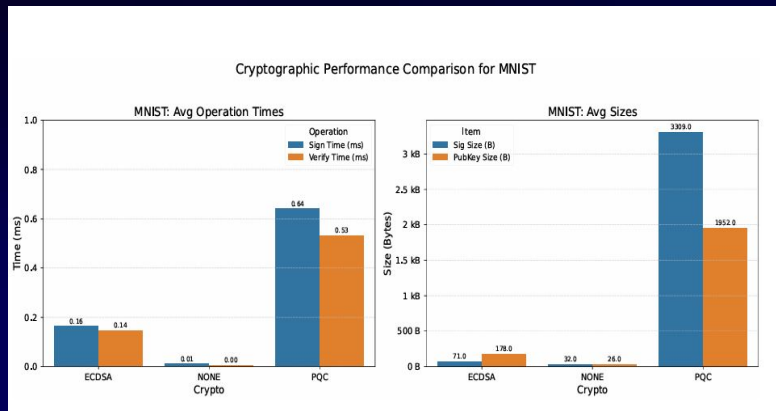
**2. Robust Model Architecture:** The CreditRiskNet model combines categorical features with nn and numerical features with BatchNorm1d. Layers are embedded. This architecture is made to efficiently manage a variety of data types.

The model can learn meaningful representations of discrete categories through the use of embedding layers for categorical features. This can be important for generalizing across different client datasets, even if the distribution of categories varies.

**3. The Generalization Capability of Federated Averaging:** The LoggingStrategy builds upon the fundamental Federated Averaging (FedAvg) algorithm, which is naturally robust to a certain extent of non-IIDness. FedAvg seeks to identify a global model that performs well across the aggregated distribution by averaging the model updates from several clients, even if their local data differs.

For moderately non-IID data (particularly if pre-processed FedAvg can still achieve good convergence, even though extreme non-IIDness can still cause problems.

# RESULTS



- **Observed Changes:**

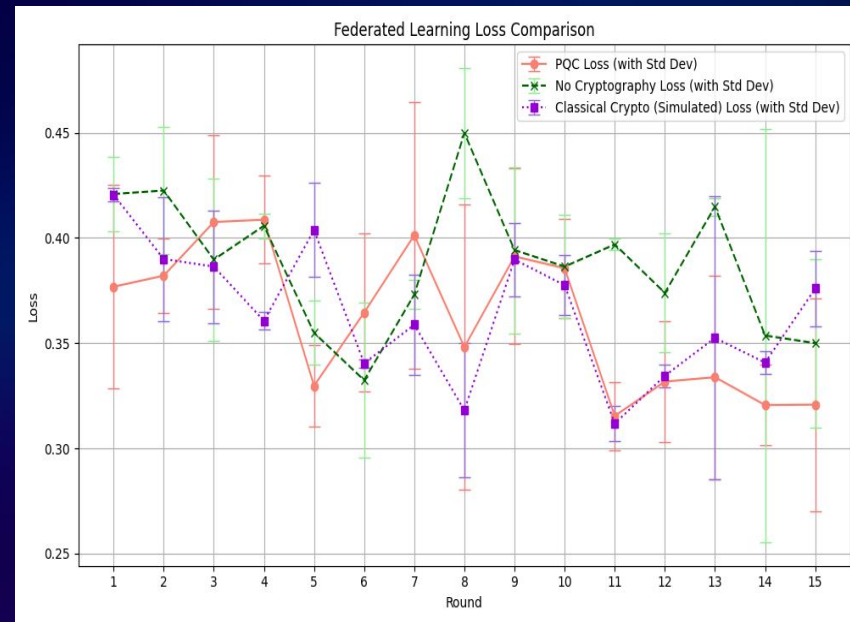
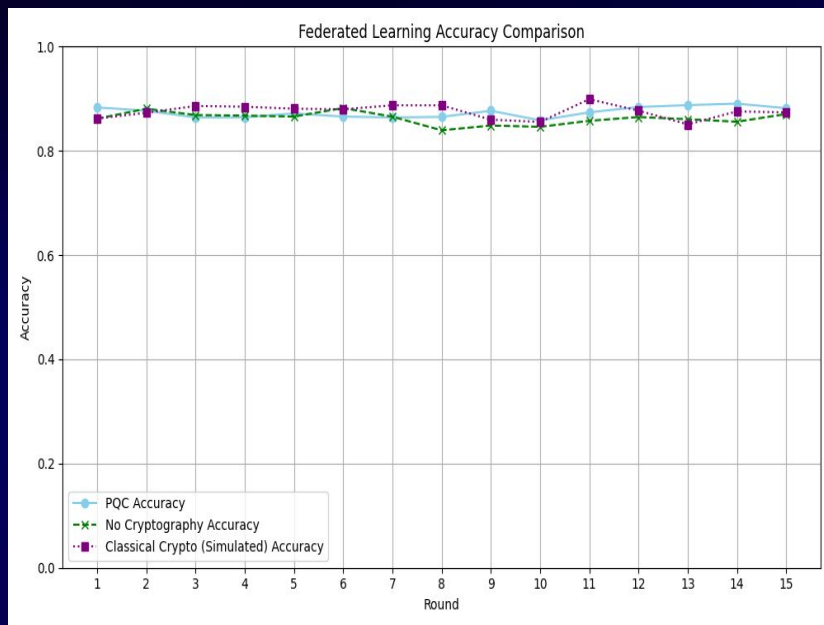
- "Signature: Key and Signature Sizes" clearly shows that Post-Quantum Cryptography (PQC) algorithms (Dilithium2, Falcon-512) have **significantly larger key and signature sizes** compared to classical ECDSA-P256.
- "Signature: Performance" highlights that PQC algorithms also exhibit **higher latency for signing and verification** operations than classical methods.

- **Our Unique Contribution & Key Takeaway:**

- We **strategically use Falcon-512 for clients with more data and Dilithium2 for clients with less data** to optimize the impact of these larger sizes and higher latencies.
- This intelligent adaptation allows our framework to integrate quantum-safe security into a **Federated Learning setup with a challenging non-IID credit risk dataset**, demonstrating that the *overall system performance* (as seen in later plots) remains robust despite these fundamental cryptographic overheads.



# RESULTS

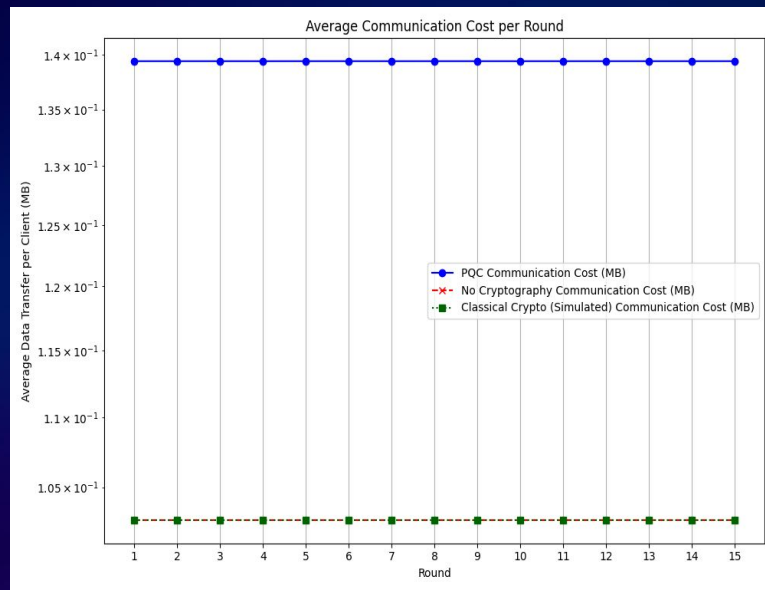
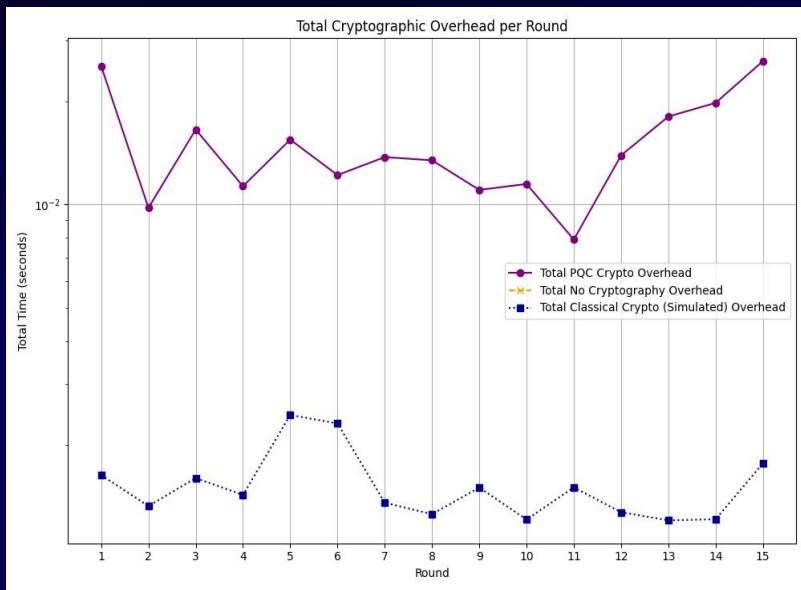


**Our PQC-enabled Federated Learning framework achieves model performance (accuracy and loss) that is strikingly similar to, and often indistinguishable from, scenarios with no cryptography or simulated classical cryptography.**

**In summary:** The accuracy and loss plots confirm that our PQC-enabled Federated Learning model successfully integrates quantum-resistant cryptography **without compromising the core machine learning performance**. This is a critical validation, demonstrating that we can achieve future-proof security for sensitive applications like credit risk analysis while maintaining high model utility.



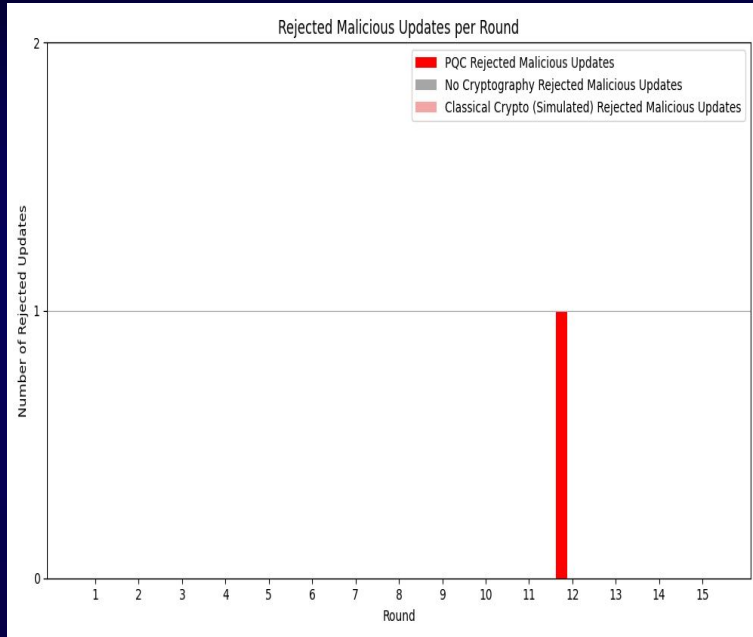
# RESULTS



Our PQC-enabled FL introduces **measurable and substantial increases** in both **computational overhead** and **communication cost** per round compared to non-cryptographic and simulated classical approaches. This is due to the inherently larger sizes and greater complexity of quantum-safe cryptographic operations.

**Uniqueness:** We uniquely quantify these trade-offs within a **decentralized, hybrid PQC-FL framework** that **adaptively optimizes signature algorithms**, providing a comprehensive benchmark for future-proof federated learning.

# RESULTS



Key takeaway: This is a direct demonstration of our framework's security. We can observe that only the PQC-enabled scenario shows rejected malicious updates (specifically, the red bar at round 12 in the example plot, if your malicious client was active and detected). The "No Cryptography" and "Classical Crypto (Simulated)" scenarios should show zero rejections, as they lack the cryptographic mechanisms (digital signatures and their verification) to detect tampering or impersonation. This plot directly supports your claim of enhanced security against MITM attacks.

# Conclusion

**Introduction:** Our Decentralized PQC-FL framework provides quantum-resistant security for sensitive AI, like credit risk analysis.

**Uncompromised Model Utility:** PQC integration does not degrade model accuracy or increase loss, confirming its non-invasive nature.

**Robust Quantum-Safe Security:** The framework ensures confidentiality, integrity, and authenticity, successfully rejecting malicious, tampered updates.

**Quantified & Managed Trade-offs:** PQC introduces measurable overheads in computation and communication, which are optimized by our adaptive signature strategy.

**Empirical Validation:** Rigorous testing on non-IID credit risk data confirms PQC-FL's viability and performance in real-world settings.

**In essence:** Our PQC-FL model is a significant step towards a future-proof, secure, and privacy-preserving AI ecosystem.

# FUTURE SCOPE

- 1. Optimize PQC Performance:** Explore hardware acceleration and dynamic algorithm selection based on network/compute.
- 2. Strengthen Privacy:** Integrate Differential Privacy or Homomorphic Encryption for enhanced data protection.
- 3. Improve Robustness:** Add Byzantine fault tolerance and advanced malicious client detection.
- 4. Enhance Scalability:** Implement dynamic server management and explore peer-to-peer FL.
- 5. Expand Applicability:** Support more complex models and diverse data types.

THANK YOU